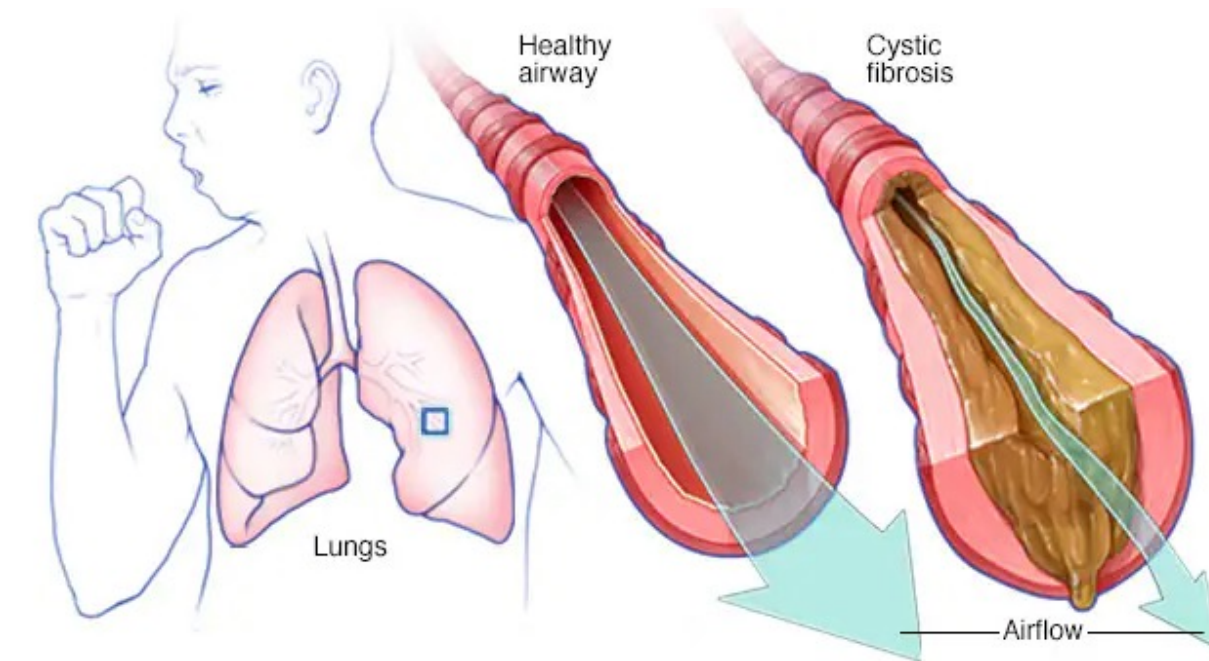


Project – Main Assignment

Practical information

- ❑ **Time:** Every afternoon, from ~15:00 to 17:00
- ❑ **Duration:** Throughout the entire week
- ❑ Apply your knowledge to tackle a real-world problem at a larger scale than the exercises
- ❑ Work on your own or in groups
- ❑ TAs are available for questions
- ❑ **Not mandatory** but highly recommended
- ❑ Solutions will be published on **Friday** after the lectures

Background



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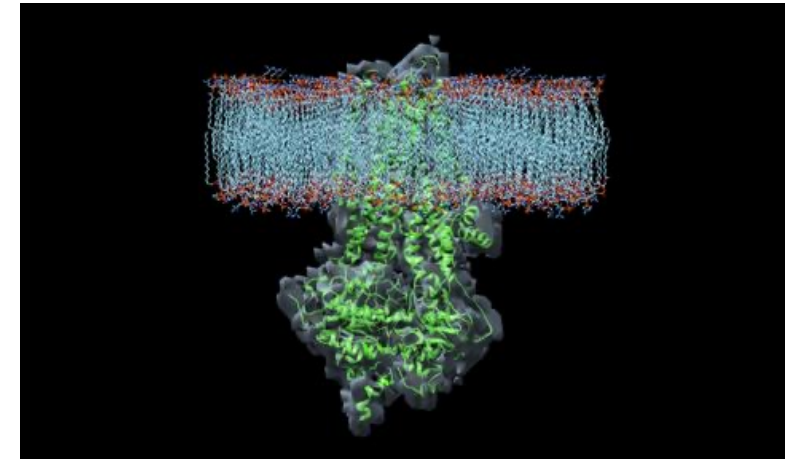
source: mayoclinic.org

Cystic fibrosis (CF)

- Genetically inherited disease
- Produces thick and sticky mucus in organs, including the lungs and pancreas
- Clogs the airways and makes it difficult for patients to breathe
- No cure is available; only symptom management, such as airway clearance

Genomic facts of Cystic Fibrosis

- CF is caused by mutations in Cystic Fibrosis Transmembrane Conductance Regulator (CFTR).
- The CFTR is an ion channel protein, acting like gates in a cell membrane that control the traffic of molecules through the membrane.
- For regular people, CFTR acts as a gate for chloride ions. When chloride leaves the cell, it carries water with it, which makes mucus less thick.
- For patients with CF, gene mutations in CFTR prevent this functionality, causing the mucus stays sticky and thick.



source: cff.org

More about the CFTR gene

- The **CFTR gene** is located on **chromosome 7** in the human genome
- More than 1,500 mutations are known to cause cystic fibrosis (CF)
- One important class of mutations is non-synonymous mutations that introduce a **premature termination codon (PTC)**, resulting in a truncated (shortened) CFTR protein
- **We are particularly interested in mutations that create these premature stop signals**, because they prevent the CFTR protein from being fully produced

Goal of the project

Write a Python program that:

- Extract the correct CFTR transcript from the human genome
- Translate it into its corresponding amino acid sequence
- Determine if one or more patients have a premature stop codon, that is, **which shortened amino acid sequence**

You will be guided step by step towards the final goal

For more details: Canvas -> Modules -> Project

Data

Filename	Description
Homo_sapiens.GRCh38.93.gtf	Human reference annotation file (GTF format)
Homo_sapiens.GRCh38.dna_sm.chromosome.7.fa	Human reference DNA sequence for chromosome 7 (fasta format)
Patient1.fa	DNA sequence data of chr 7 for Patient 1 (fasta format)
Patient2.fa	DNA sequence data of chr 7 for Patient 2 (fasta format)
Patient3.fa	DNA sequence data of chr 7 for Patient 3 (fasta format)
Patient4.fa	DNA sequence data of chr 7 for Patient 4 (fasta format)
Patient5.fa	DNA sequence data of chr 7 for Patient 5 (fasta format)

Fasta format

```
>MT dna:chromosome chromosome:GRCh38:MT:1:16569:1 REF
GATCACAGGTCTATCACCTTATTAACCACTCACGGGAGCTCTCCATGCATTTGGTATTTT
CGTCTGGGGGGGTATGCACGCGATAGCATTGCGAGACGCTGGAGCCGGAGCACCTTATGTC
GCAGTATCTGTCTTTGATTCCTGCCTCATCCTATTATTTATCGCACCTACGTTCAATATT
ACAGGCGAACATACTTACTAAAGTGTGTTAATTAATTAATGCTTGTAGGACATAATAATA
ACAATTGAATGTCTGCACAGCCACTTTCCACACAGACATCATAACAAAAAATTTCCACCA
AACCCCCCCTCCCCCGCTTCTGGCCACAGCACTTAAACACATCTCTGCCAAACCCCAAAA
ACAAAGAACCCTAACACCAGCCTAACCAGATTTCAAATTTTATCTTTTGGCGGTATGCAC
TTTTAACAGTCACCCCCCAACTAACACATTATTTTCCCCTCCCCTCCCATACTACTAAT
CTCATCAATACAACCCCCCGCCCATCCTACCCAGCACACACACACACCGCTGCTAACCCCA
CCCCGAACCAACCAAACCCCAAAGACACCCCCCACAGTTTATGTAGCTTACCTCCTCAA
```

GTF format

- GTF stands for Gene transfer format
- Holds information about the gene structure
- Tab-delimited, i.e., columns are separated by tabs ('\t')

Columns of GTF file

<seqname> <source> <feature> <start> <end> <score> <strand> <frame> [attributes]

1. **<seqname>**: The name of the sequence (typically a chromosome number).
2. **<source>**: The source of the annotation (e.g., ENSEMBL).
3. **<feature>**: The type of feature (e.g., gene, transcript, exon).
4. **<start>**: The starting position of the feature in the sequence, **starting from 1**.
5. **<end>**: The ending position of the feature in the sequence, **end position included**.
6. **<score>**: A score between 0 and 1000, or . if not applicable (indicating the reliability of the annotation).
7. **<strand>**: The strand on which the feature is located (`+` for the forward strand, `-` for the reverse strand). (**only forward strand in this project**)
8. **<frame>**: The reading frame, one of '0', '1' or '2', or `.` if not applicable.
9. **[attribute]**: A list of key-value pairs providing additional information about the feature.

Attribute of GTF

- **A semicolon (;)-separated list of key-value pairs**
- For each key-value pair, key is one word, and the value is quoted by double quotes, which may contain multiple words
- A key can be repeated multiple times.

Some attributes (always semi-colon separated key-value pairs):

- `gene_id`: The stable identifier for the gene
- `gene_version`: The stable identifier version for the gene
- `gene_name`: The official symbol of this gene
- `gene_source`: The annotation source for this gene
- `transcript_id`: The stable identifier for this transcript
- `transcript_name`: The symbol for this transcript derived from the gene name
- `exon_id`: The stable identifier for this exon

GTF example

<seqname> [attributes]	<source>	<feature>	<start>	<end>	<score>	<strand>	<frame>
---------------------------	----------	-----------	---------	-------	---------	----------	---------

```
1 havana gene 11869 14409 . + . gene_id "ENSG00000223972"; gene_version "5";  
gene_name "DDX11L1"; gene_source "havana"; gene_biotype  
"transcribed_unprocessed_pseudogene";
```

```
1 havana transcript 11869 14409 . + . gene_id "ENSG00000223972"; gene_version  
"5"; transcript_id "ENST00000456328"; transcript_version "2"; gene_name  
"DDX11L1"; gene_source "havana"; gene_biotype  
"transcribed_unprocessed_pseudogene"; transcript_name "DDX11L1-202";  
transcript_source "havana"; transcript_biotype "processed_transcript"; tag  
"basic"; transcript_support_level "1";
```

```
1 havana exon 11869 12227 . + . gene_id "ENSG00000223972"; gene_version "5";  
transcript_id "ENST00000456328"; transcript_version "2"; exon_number "1";  
gene_name "DDX11L1"; gene_source "havana"; gene_biotype  
"transcribed_unprocessed_pseudogene"; transcript_name "DDX11L1-202";  
transcript_source "havana"; transcript_biotype "processed_transcript";  
exon_id "ENSE00002234944"; exon_version "1"; tag "basic";  
transcript_support_level "1";
```

How to get DNA sequence of a gene

```
<seqname> <source> <feature> <start> <end> <score> <strand> <frame> [attributes]
```

```
1  havana  gene  11869  14409  .  +  .  gene_id  "ENSG00000223972";  gene_version  "5";  gene_name  "DDX11L1";  
gene_source  "havana";  gene_biotype  "transcribed_unprocessed_pseudogene";
```

- Get full DNA sequence of human ch7 from the file
"Homo_sapiens.GRCh38.dna_sm.chromosome.7.fa"
- The position is 1-based, with both ends included

```
start_pos = 11869
```

```
end_pos = 14409
```

```
seq_gene = ch7_dna_sequence[start_pos - 1:end_pos]
```

Getting started

- Create a folder called **project**
- Download and extract the project files in this folder (download link is provided in the project page)
- Work with Jupyter or any text editor (e.g., Spyder, Sublime)
- Ask questions if something is unclear
- Speak to a TA or discuss with your neighbour
- Or use the discussion section in Canvas
- Find out more info at Canvas -> Modules -> Project

Questions?